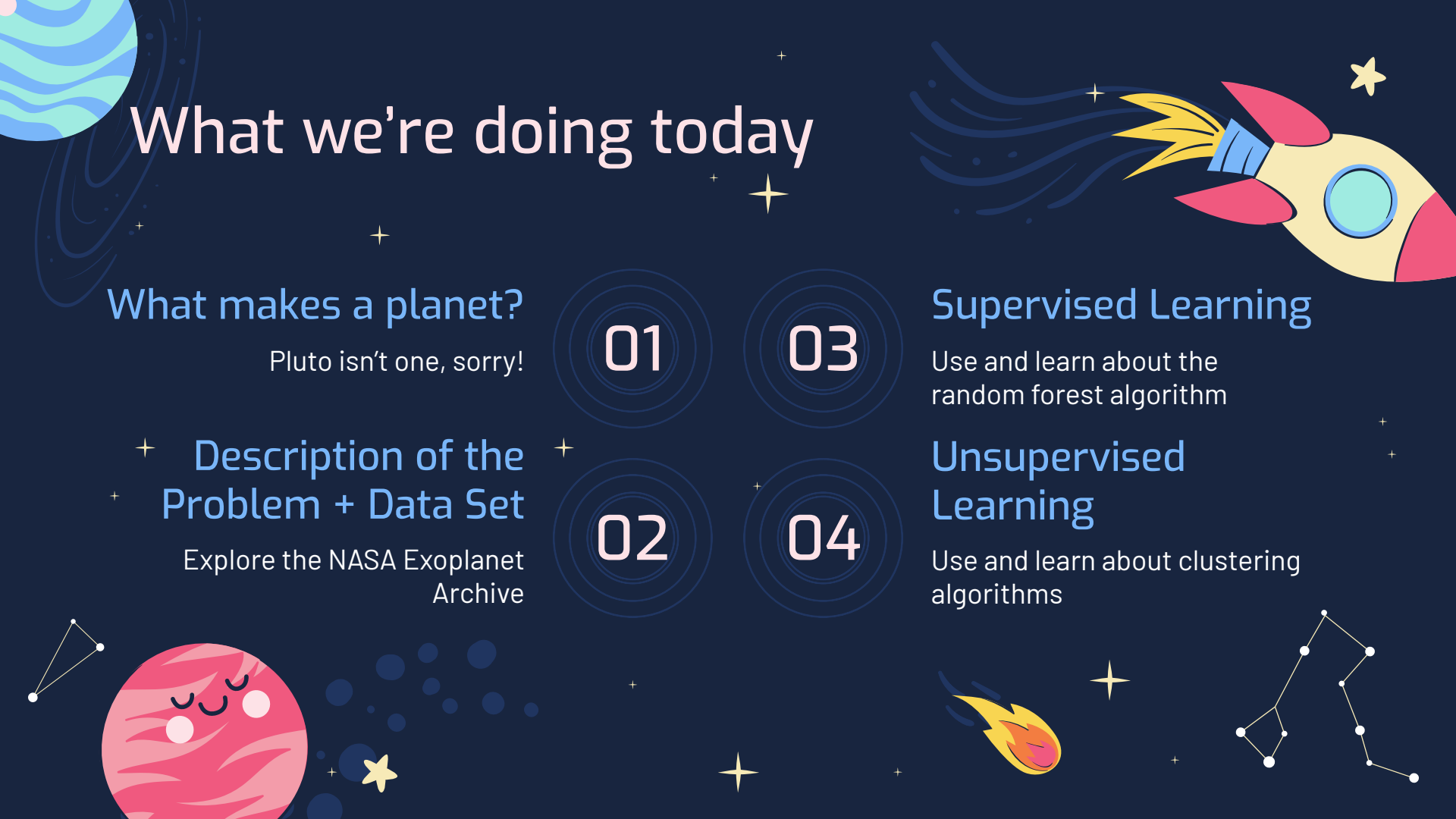




Hunting for Exoplanets

Investigating Kepler Objects of Interest





What we're doing today

What makes a planet?

Pluto isn't one, sorry!

Description of the Problem + Data Set

Explore the NASA Exoplanet Archive

01

03

Supervised Learning

Use and learn about the random forest algorithm

02

04

Unsupervised Learning

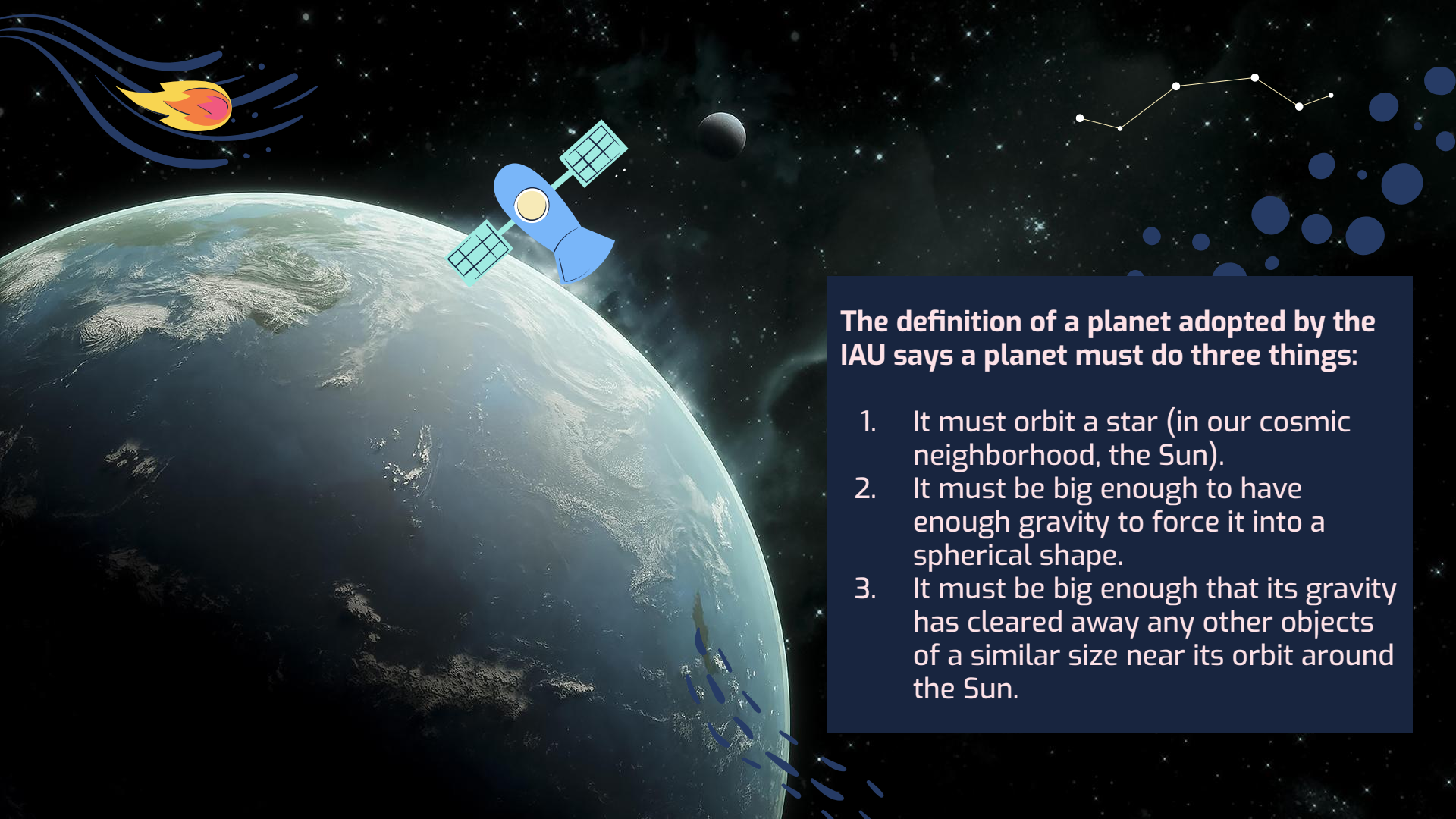
Use and learn about clustering algorithms

The task

Data from several different space-based exoplanet surveying missions have enabled discovery of thousands of new planets outside our solar system, but most of these exoplanets were identified manually. With advances in artificial intelligence and machine learning (AI/ML), it is possible to automatically analyze large sets of data collected by these missions to identify exoplanets. Your challenge is to create an AI/ML model that is trained on one or more of the open-source exoplanet datasets offered by NASA and that can analyze new data to accurately identify exoplanets.

- NASA Space Apps '25





The definition of a planet adopted by the IAU says a planet must do three things:

1. It must orbit a star (in our cosmic neighborhood, the Sun).
2. It must be big enough to have enough gravity to force it into a spherical shape.
3. It must be big enough that its gravity has cleared away any other objects of a similar size near its orbit around the Sun.

The NASA Exoplanet Archive:

- What is a KOI?
- Features of Interest
- Target Variable
Dataset



Feature Selection

Koi_period [days]

Orbital period - Kepler's
third law of motion

Koi_srad [Solar radii]

Stellar Radius

Koi_duration[hrs]

Transit Duration. Related to
orbital speed and stellar
radius

koi_steff[K]

Stellar Effective
Temperature

koi_depth [ppm]

Transit Depth

Koi_prad [Earth radii]

Planetary radius

koi_teq[K]

Equilibrium
Temperature

Data Splitting

Training Set

- Approx. 70-80%
- Learning patterns

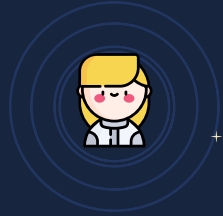
Testing Set

- Approx. 20-30%
- Keep this separate

Beware of Overfitting: creating a model that matches (memorizes) the training set so closely that the model fails to make correct predictions on new data

Beware of Underfitting: occurs when a model is too simple, which can be a result of a model needing more training time, more input features, or less regularisation.

Supervised vs Unsupervised Learning



Supervised

Produces accurate and reliable models with high-quality data.
Easy to evaluate performance due to known outcomes.
Can handle various tasks, including classification and regression.
Effective for applications that predict future trends using historical data.

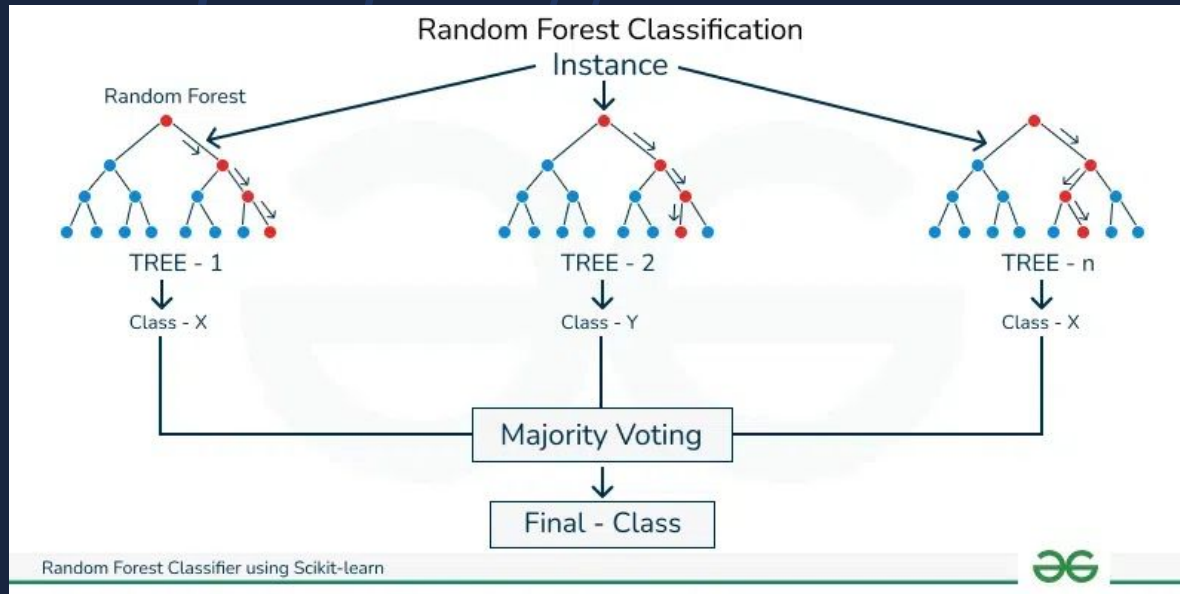


Unsupervised

Excels in exploring raw, unstructured data to uncover hidden patterns.
Handles large volumes of data efficiently.
Simplifies complex, high-dimensional data quickly.
Fast classification of data.



Supervised algorithms: Random Forests





Unsupervised algorithms: Clustering

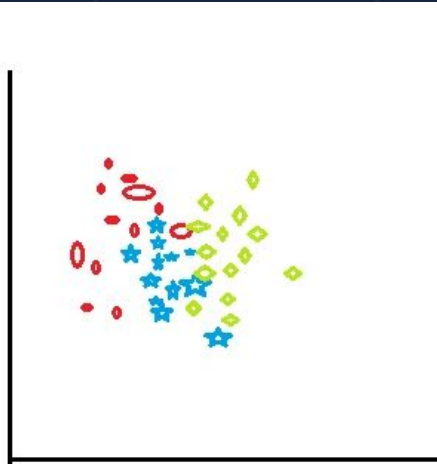


fig 1: before applying
k-means clustering

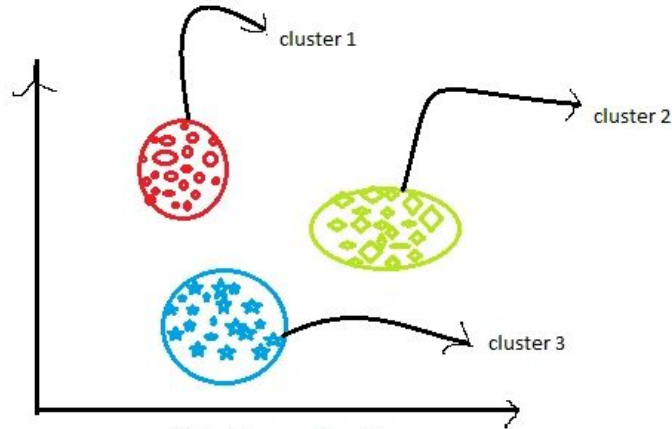


fig 2: After applying K-
means clustering

